

TO: KPC Revenue Assurance Stakeholders & PLP Supervisors

FROM: Null_Terminators Team

SUBJECT: Problem-Framing: Linking O2C Data to KPC Revenue Leakage

The Problem

Kenya Pipeline Company (KPC) is experiencing critical revenue leakage within its Order-to-Cash (O2C) cycle. Because operational data is currently siloed, KPC is vulnerable to unmetered product loading, invoicing/tariff errors, underpayments, and organized fraud (e.g., product substitution and depot-shopping). Manual auditing cannot catch these multi-party discrepancies.

Linking the Data to the Leakage Points

To solve this, we must unify KPC's fragmented datasets and map them directly to the specific vulnerabilities in the O2C cycle:

- **Detecting Unmetered Loading (Physical vs. Commercial):** By reconciling physical gantry ground truth (`depot_loading_logs.csv`, `depot_daily_inventory.csv`) against commercial waybills (`dispatches.csv`), we expose physical fuel movement that lacks commercial documentation.
- **Detecting Billing & Payment Leaks (Commercial vs. Financial):** By matching financial settlement records (`invoices.csv`, `payments.csv`) against the commercial dispatches, we immediately isolate missing invoices, misapplied tariffs, split installments, and underpayments.
- **Detecting Network Fraud (Transactions vs. Master Data):** By strictly validating all movement against versioned reference data (`omcs.csv`, `products.csv`, `depots.csv`, `tariffs.csv`), we establish the baseline needed to flag anomalies like product swapping and OMC collusion.

Conclusion Integrating these nine specific datasets is the mandatory first step to mapping the entire O2C lifecycle. This linkage transforms fragmented data into a cohesive, automated defense against revenue loss.